

A Deep Dive Into UCle-3D

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Meet the Presenter



Zuoguo (Joe) Wu

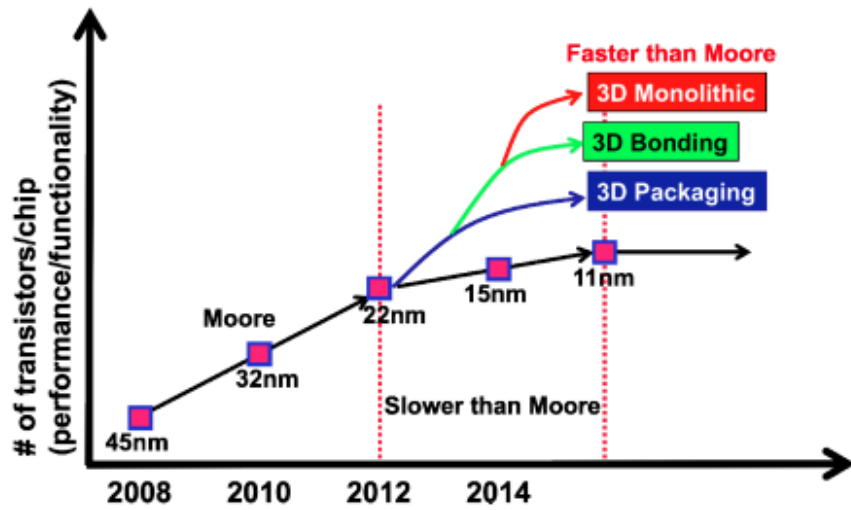
Zuoguo Wu is a Senior Principal Engineer at Intel, where he manages an I/O circuits and architecture team working on the latest interfaces. He also serves as the UCle Consortium's Electrical Working Group Co-Chair.

He is a principal author of the UCle specification and defined its PHY architecture and circuit and link analysis. He previously contributed to the PCIe spec.

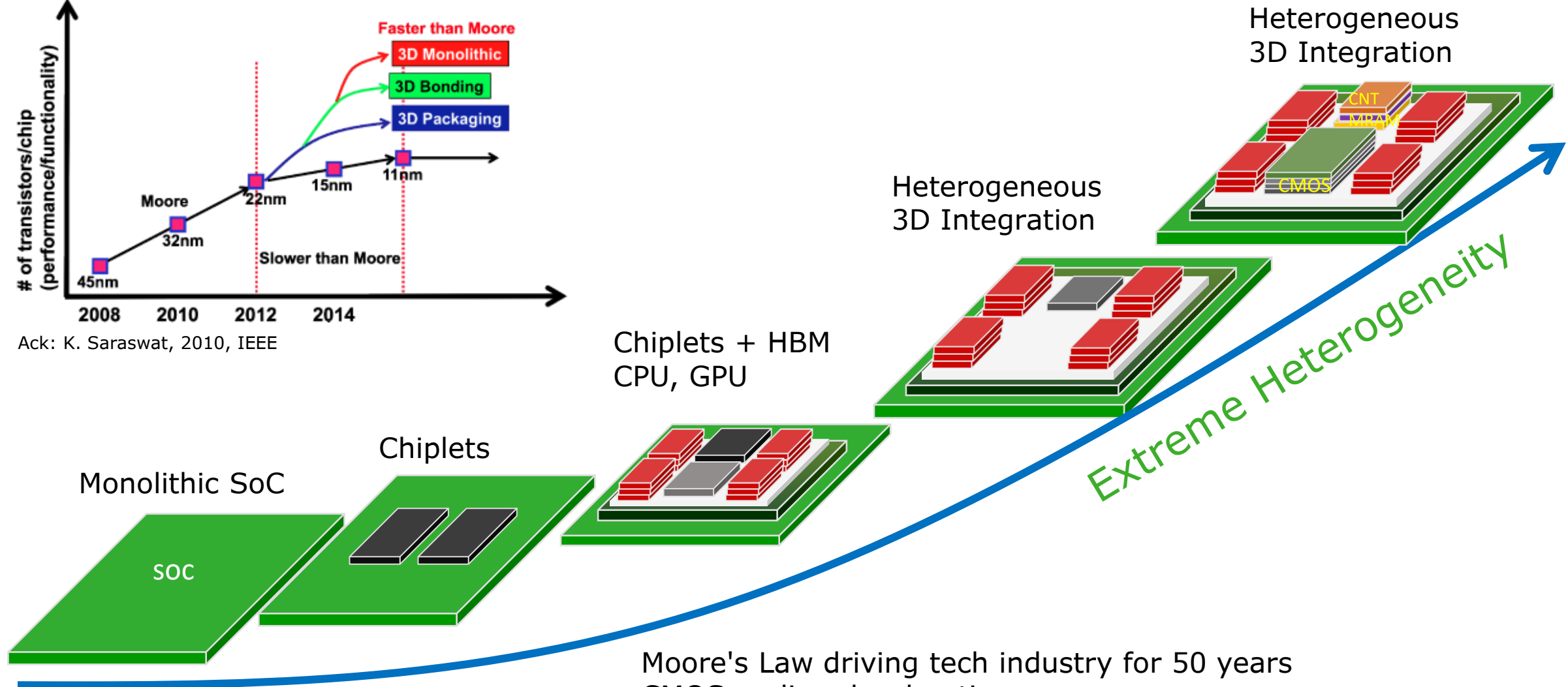
He holds 140 patents and has published over 30 external and Intel-internal papers.

He earned a PhD in electrical engineering at Texas A&M University.

Monolithic to Chiplets



Ack: K. Saraswat, 2010, IEEE



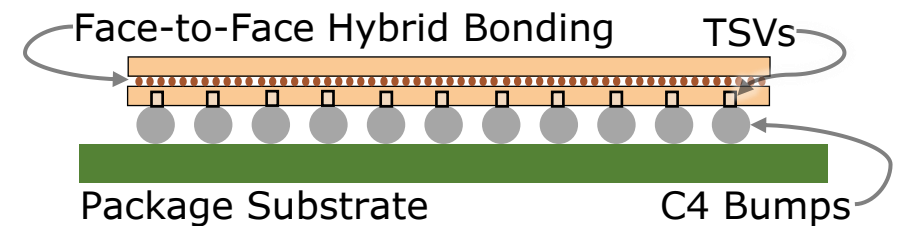
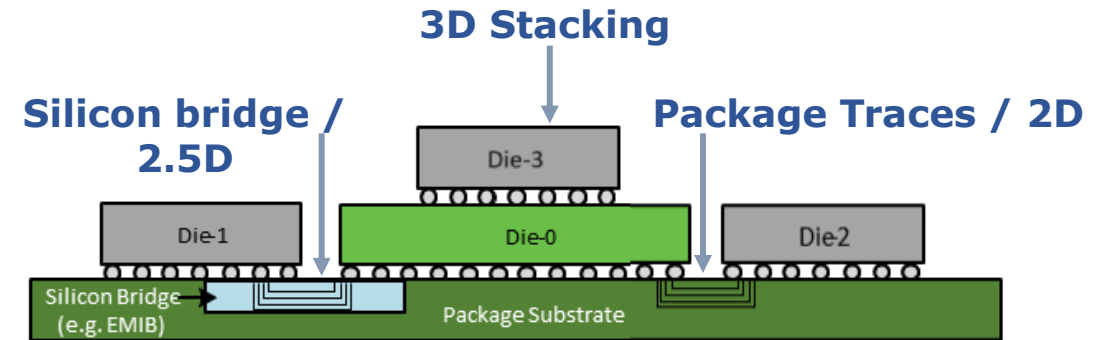
Moore's Law driving tech industry for 50 years
CMOS scaling decelerating
Industry shifting from monolithic SoC to chiplets

Ack: Stefan Rusu

Types of Die-to-Die IO

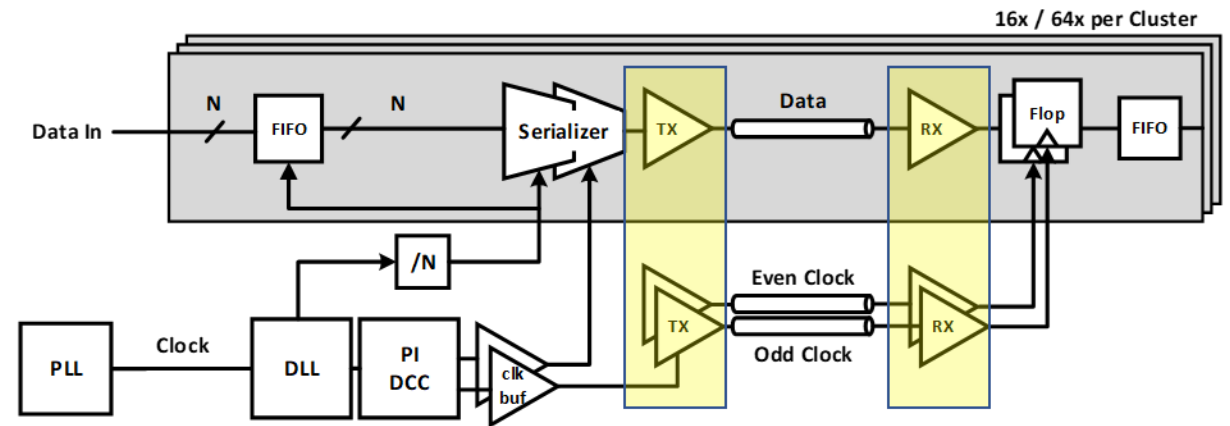
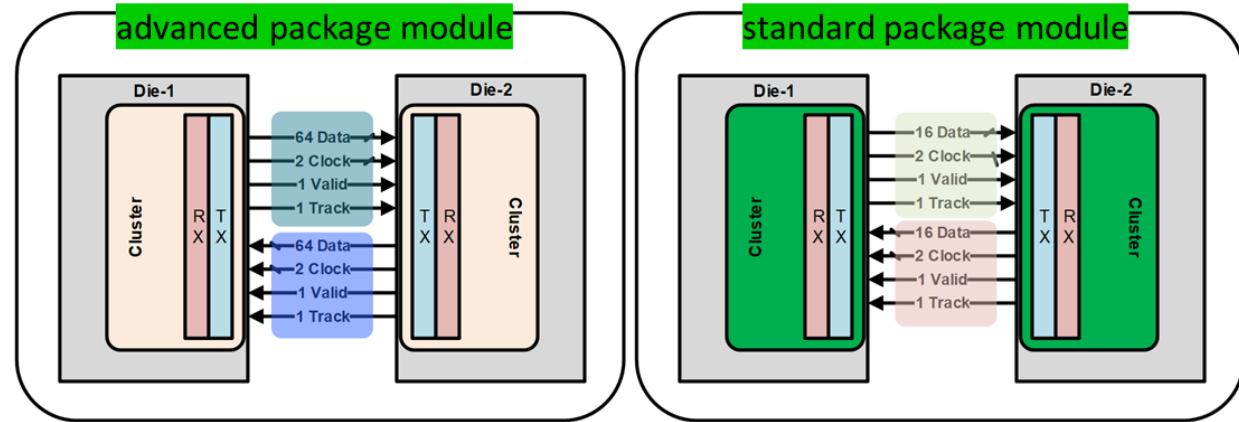
- 2D Integration (Standard package traces)
 - Best package floorplan flexibility
 - Examples: UCIE, XSR
- 2.5D Integration (Si bridge, Si Interposer)
 - Best lateral connection KPIs
 - Relatively higher package cost and constrained package floorplan
 - Examples: UCIE, HBM
- 3D Integration (Si Interposer/HBI)
 - Best power efficiency and lowest latency
 - Best bandwidth density
 - Higher package cost and constrained Silicon floorplan
 - Examples: UCIE-3D, HBM/DRAM

Packaging Technology examples for chiplets



UCIe 2D/2.5D PHY Recap

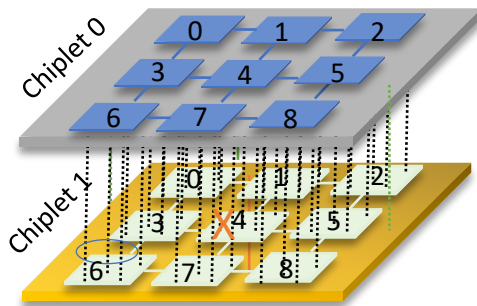
- **Low Datapath Latency:** 12 UI up to 16Gb/s, 16 UI 24-32Gb/s.
 - @16Gb/s $\leq 2\text{ns}$ when including adapter
- **Ultra Fast Idle Exit (L0-idle):** $\leq 1\text{ns}$, orders of magnitude better than typical serial I/O.
- **Valid Framing and Retimer support,** Hamming distance 4 encoding to improve reliability.
- L1 power state (options for further power down)
- **Redundancy & Repair** for Advanced Package
 - Two per 32 bits
- **Data Scrambling** to mitigate power supply noise
 - Does not require extra pin as DBI would
- **Tightly-Coupled Mode** for further optimization (when supported by channel)



Matched Circuit Arch
Matched Interconnect

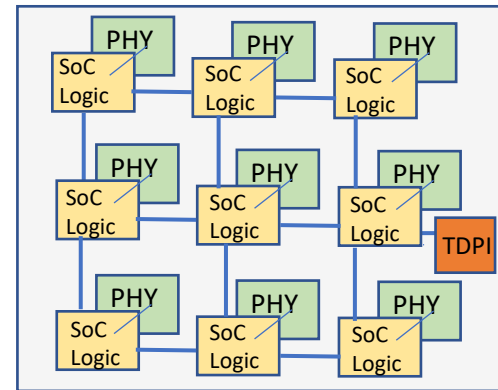
UCIe-3D Objectives / Approaches

- **Compact Design**: Circuit/logic confined to bump area; necessitates simpler circuits at lower frequencies. Example: For 1 μm bump-pitch, area per lane must be <1 sq μm .
- No D2D adapter. Low BER due to low frequency and almost 0 channel distance - No CRC/ replay is needed
- **Minimal PHY**: The SoC Logic (NoC) connects directly to the PHY
- All debug/testability hooks inside a common block across all UCIe-3D Links that is connected to the SoC Logic network inside the chiplet
- The lane repair becomes a cluster-wide repair and orchestrated by the SoC logic



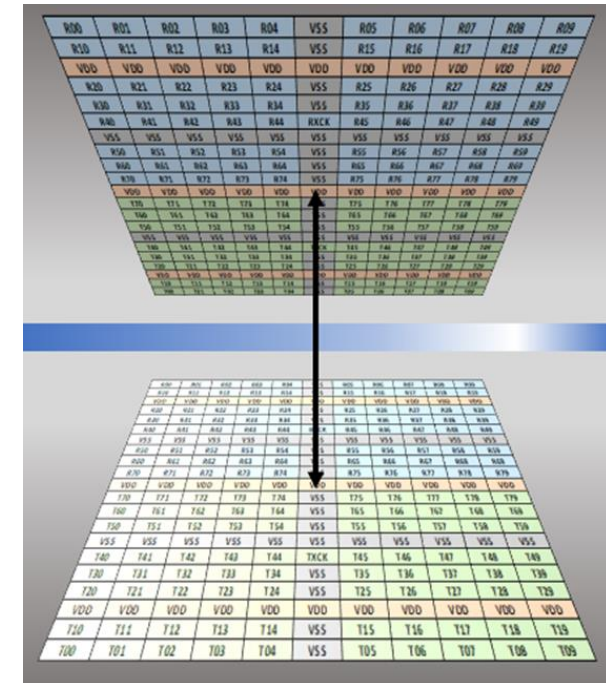
(a)

Chiplets connected across each SoC with UCIe-3D. A failure in UCIe-3D in either die results in the remaining SoCs routing around the failure



(b)

Each chiplet has its system controller logic, I/Os etc. Each SoC Logic connects to one or more UCIe-3D PHY. The common test, debug, pattern and Infrastructure (TDPI) block orchestrates Training, testing, debug etc. across chiplets.



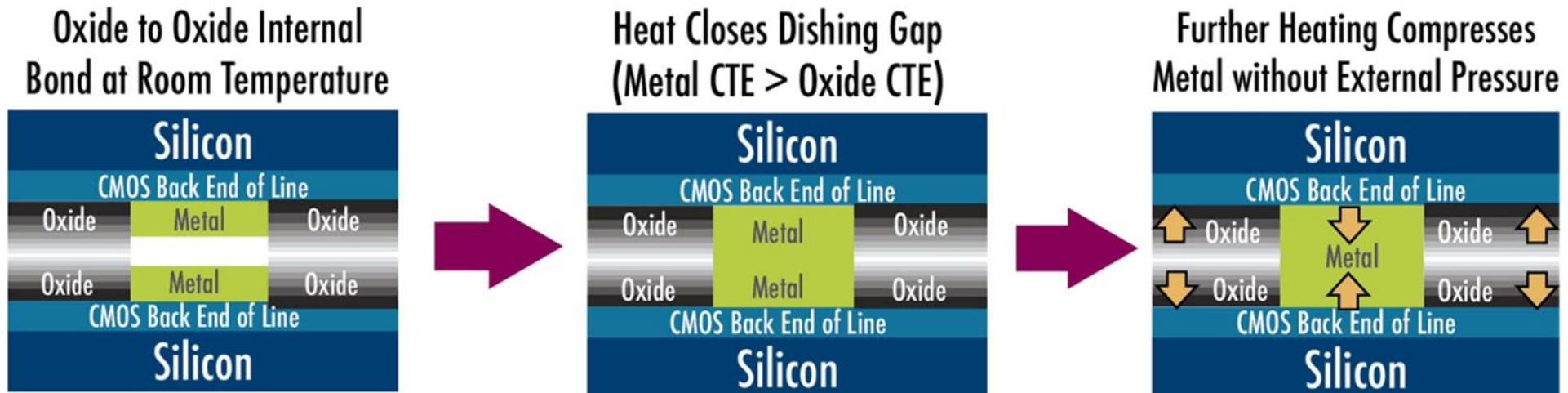
Hybrid Bonding

History of chip interconnect:

- Wire bond → Solder bump → Si Interposer / micro bump → Hybrid bonding

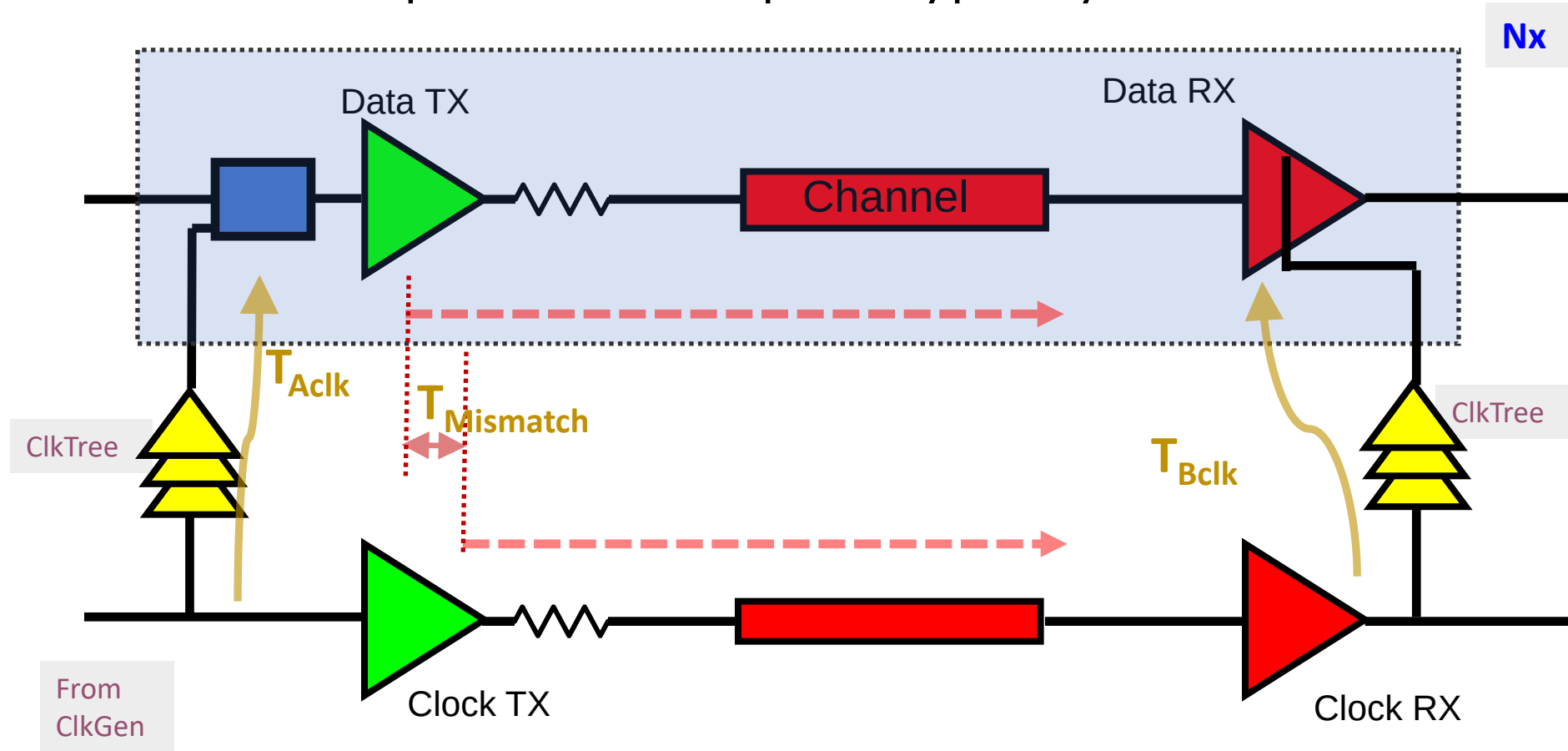
Hybrid bonding: dielectric bond combined with metal (Cu) bond

- Below 10μm bump pitch, potentially to 1μm or lower.
- Interconnect parasitics (R, C) reduced by 90% compared to solder
- Surfaces need to be extremely flat and smooth
 - A single defect can result in a lot of connection failures. Something to keep in mind for I/O repair strategy.



UCIe-3D PHY Electrical

- Achieve matched architecture benefits for noise rejection
- Avoid increased power consumption typically associated with it



$$\max(T_{Aclk}) - \min(T_{Aclk}) + \max(T_{Bclk}) - \min(T_{Bclk}) + T_{mismatch} + T_{setup} + T_{hold} < T_{cycle}$$

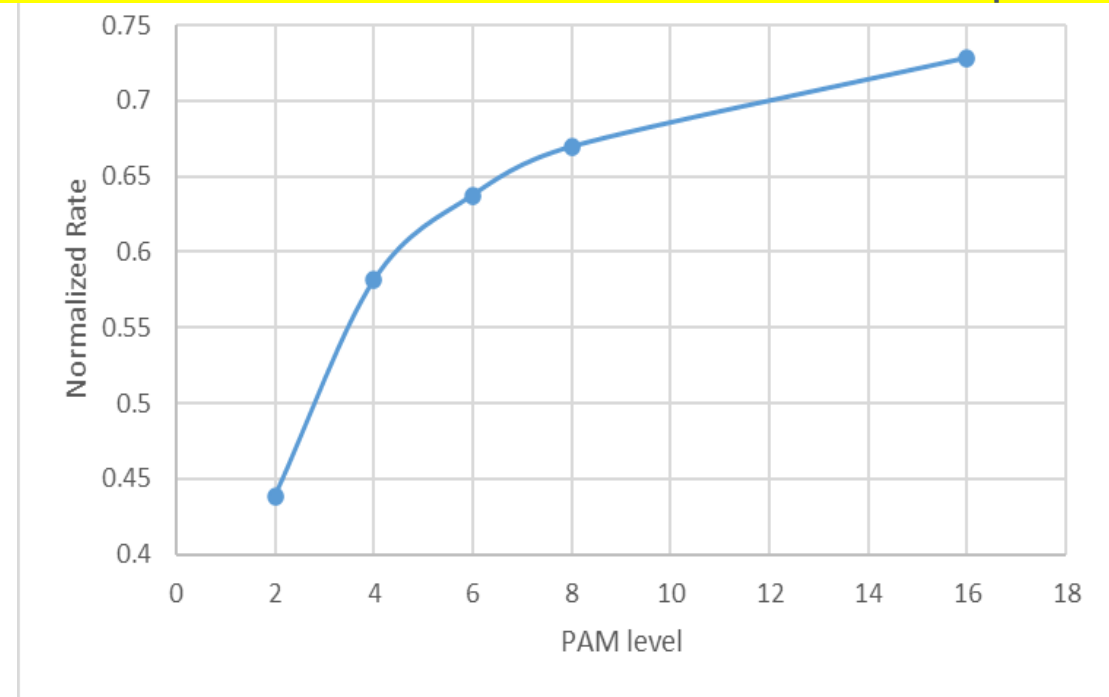
PAM-N Signaling in High-Speed I/O

- In a channel BW limited system (high loss interconnect), PAM-N can offer data rate advantages.
- PAM-4 widely used in Ethernet 50+Gb/s and PCIe 6.0/7.0, PAM-6 considered for Ethernet 224Gb/s, PAM-16 in MIPI A-PHY.
- Shannon capacity based on unlimited power and latency, arbitrary complex modulation and encoding.

Theoretical Signal-to-Noise Ration (SNR)

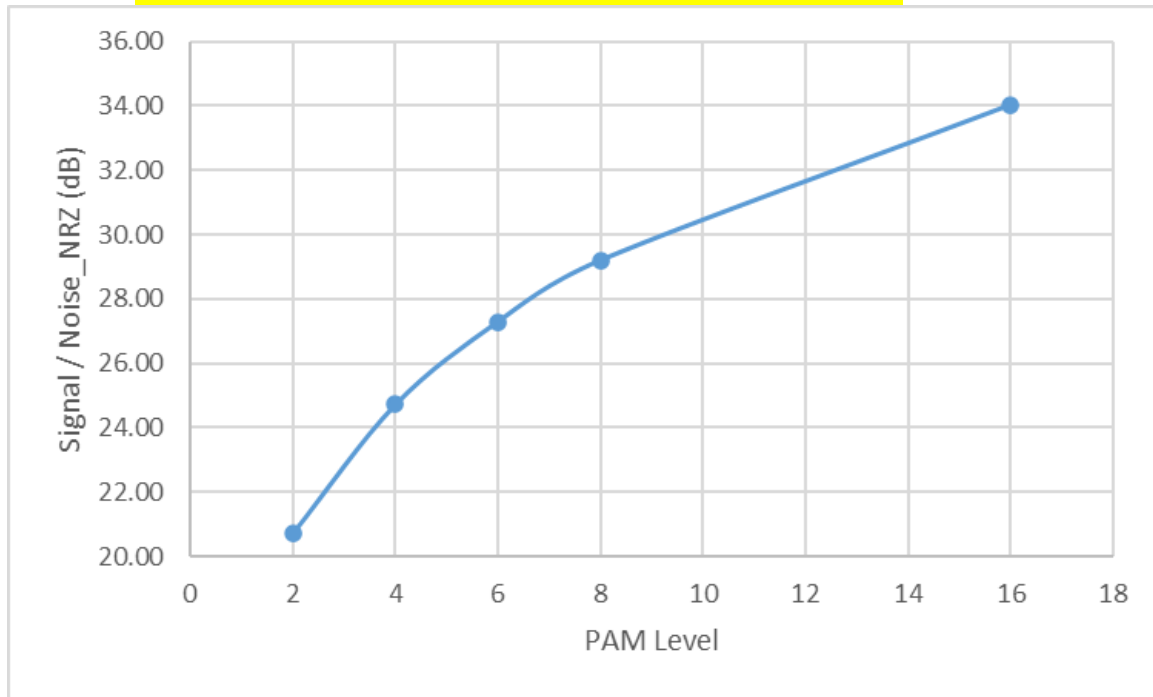
SNR	NRZ	PAM	PAM	PAM	PAM
BER	2	4	6	8	16
1.00E-03	9.80	17.12	20.88	23.47	29.59
1.00E-04	11.41	18.63	22.37	24.95	31.06
1.00E-06	13.54	20.68	24.39	26.96	33.06
1.00E-12	16.94	24.00	27.70	30.26	36.35

Ratio of Data Rate at 1E-6 vs Shannon Capacity



Power Limited System

Signal Power (BER 1E-27)
Normalized to NRZ Noise Power



- UCIe is in a power limited system, not channel BW limited.
 - From UCIe-A 45um bump pitch to UCIe 3D 9um, density scales by 25x.
 - Data rate only needs to be 1.28Gb/s to be iso-BW.
 - If data rate is the same, energy efficiency needs to be 0.01pJ/b to be iso-power.
- For a target data rate and given noise spectral density, PAM-N can reduce the Baud rate, hence total noise. But it requires much higher SNR. Total power increases.
- **NRZ is the best choice for power in a low loss system.**

Power Efficiency

- For unterminated low power I/O, dynamic power dominates. Power efficiency in pJ/b can be expressed as

$$E \left(\frac{\text{pJ}}{\text{b}} \right) = \left(\frac{C_{dat}}{4} + \frac{C_{ck}}{N} \right) V_{dd}^2$$

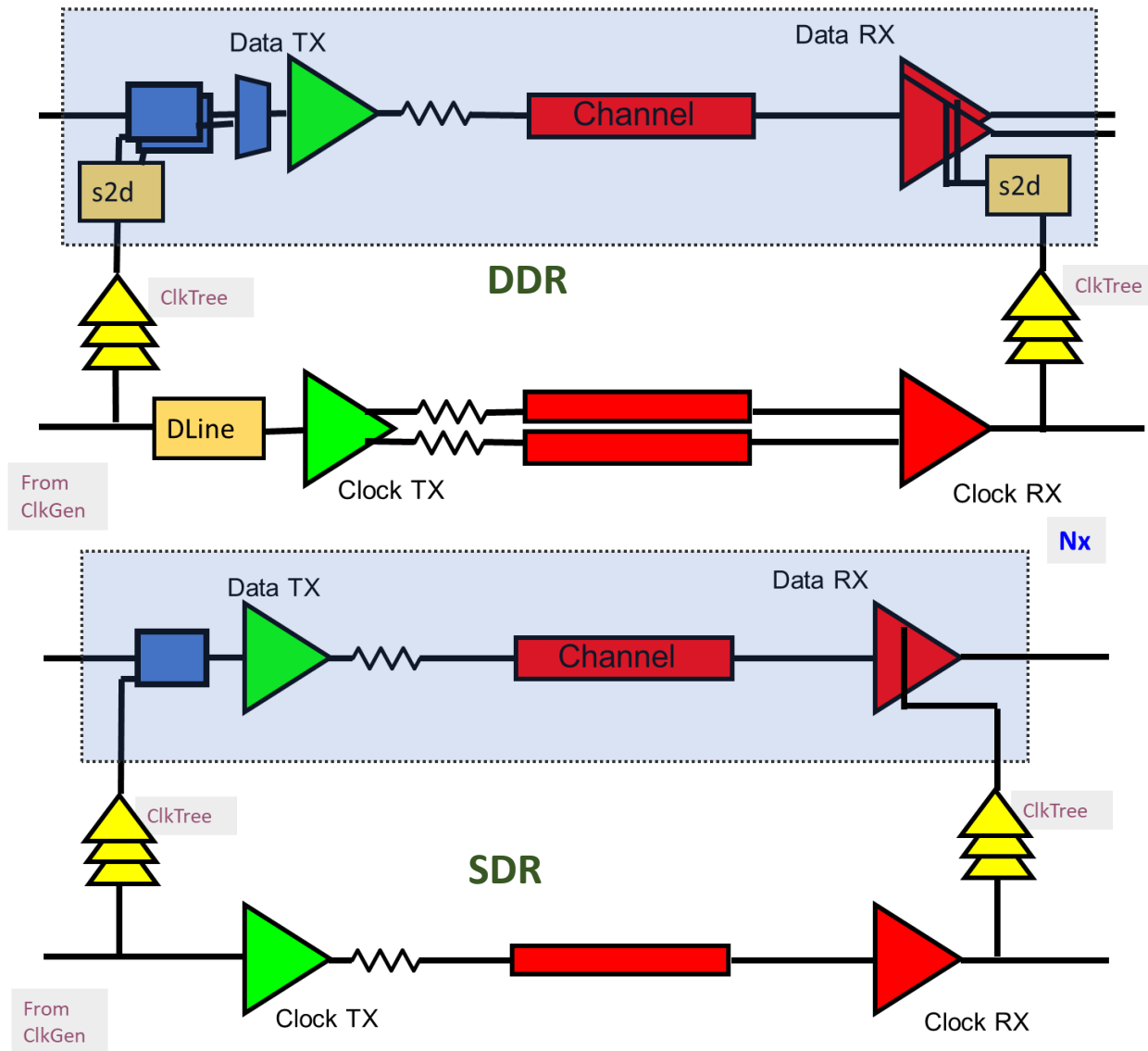
- **C_{dat}** is the total capacitance associated to data bits
- **C_{ck}** is the total capacitance associated to clock buffers, distribution and generation.
- Capacitance includes all wires and ESD
- Common circuits amortized to per data lane
- **N** is the data rate to clock frequency ratio

Optimal Power Considerations

- V_{dd} should be as low as possible
- Minimize ESD
- Aim for as few transistors as possible. Extra circuits to increase the data rate won't improve the power efficiency, since it only increases C. This is different from high-speed serial I/O.
- Uni-directional is better than bi-directional, since bi-directional bus has both TX and RX Cap at pad. Here bi-directional refers to turn-around bus, not SBD.
- Higher N improves power efficiency only if the circuit complexity increases by a factor smaller than N.
 - N-way interleaving by duplicating the circuit N times does not improve efficiency. It relaxed timing requirement, but that's not the topic here.
 - Practically $N = 1$ or 2 , since generating 4+ phases locally requires a lot of circuits such as DLL.

Clocking Trade-off

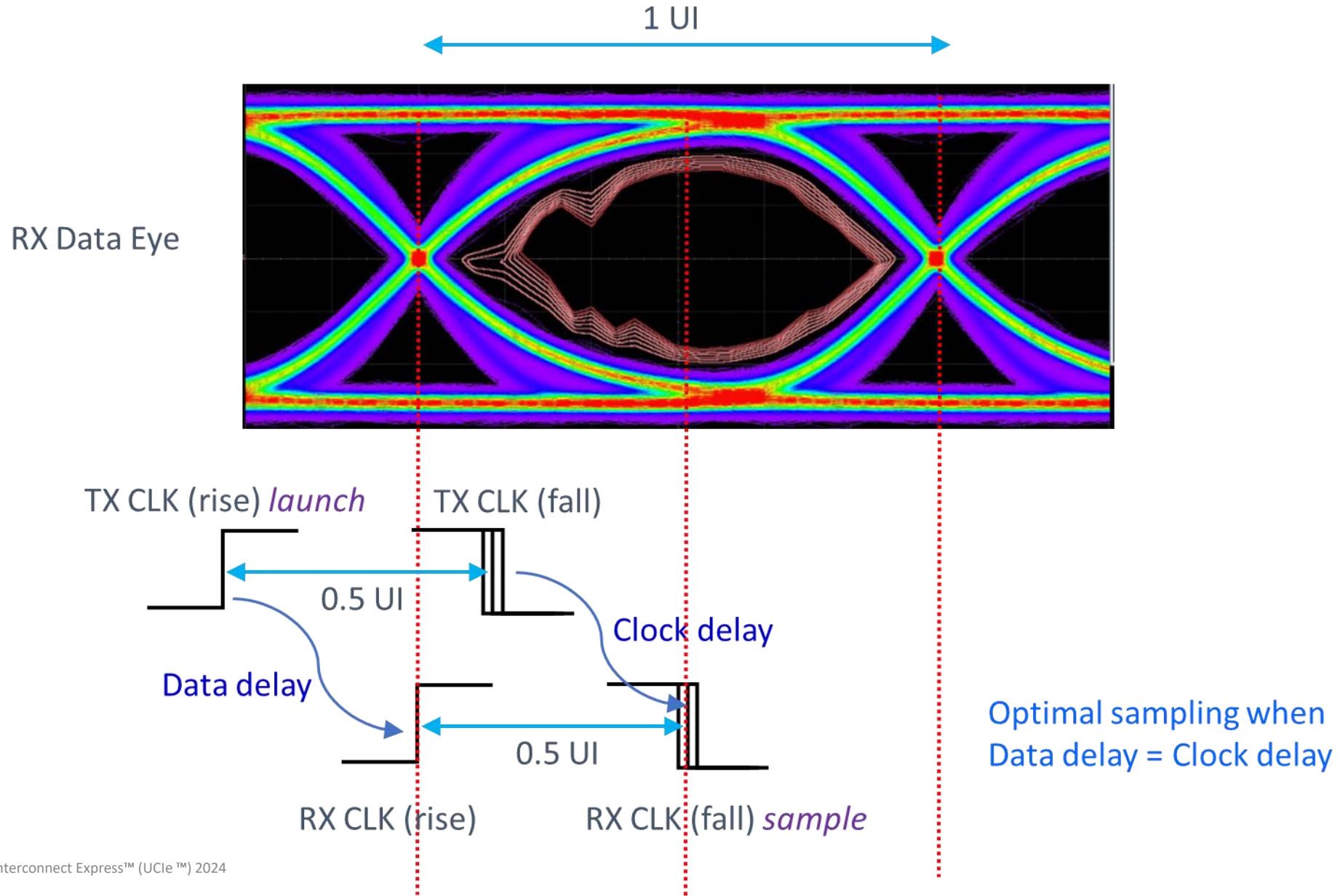
Nx



- Advantages of DDR:
 - Possible Lower Power
 - More circuits but at lower clock speed
- Advantages of SDR:
 - Better scaling to smaller bump pitch
 - Data rate change is straightforward, no need to re-calibration. Besides 4Gb/s, need to support 2, 1, 0.5...
 - Dynamic voltage and frequency scaling (DVFS)

UCIe-3D: SDR for best overall flexibility, performance and forward scaling

Launch Edge and Sample Edge



Timing and Mismatch Spec

	Name	Min	Typ	Max	Unit	UI = 250 ps @ 4Gb/s	Note
Eye Closure due to Channel	Ch			0.1	UI	25 ps	(1)
Pulse Width Deviation from 50% Clock Period	Jpw			0.08	UI pkpk	20 ps	
TX Lane-to-Lane Skew	Stx			0.12	UI pkpk	30 ps	
RX Lane-to-Lane Skew	Srx			0.12	UI pkpk	30 ps	
TX Data/Clock Differential Delay	Dtx	D_{tx_min}	D_{tx_typ}	D_{tx_max}		max – min = 50 ps	(2)
RX Data/Clock Differential Delay	Drx	D_{rx_min}	D_{rx_typ}	D_{rx_max}		max – min = 50 ps	
Alpha Factor (TX and RX)	α_{tx}			1.5			(3)
Vcc noise	nvcc			10	% pkpk		(4)
TX Data/Clock Differential RJ	Jrtx			0.05	UI pkpk at BER	12.5 ps	
RX Data/Clock Differential RJ	Jrrx			0.05	UI pkpk at BER	12.5 ps	
Sampling Aperture	Ap			0.03	UI	7.5 ps	

(1) Eye closure due to channel includes inter-symbol interference (ISI) and crosstalk

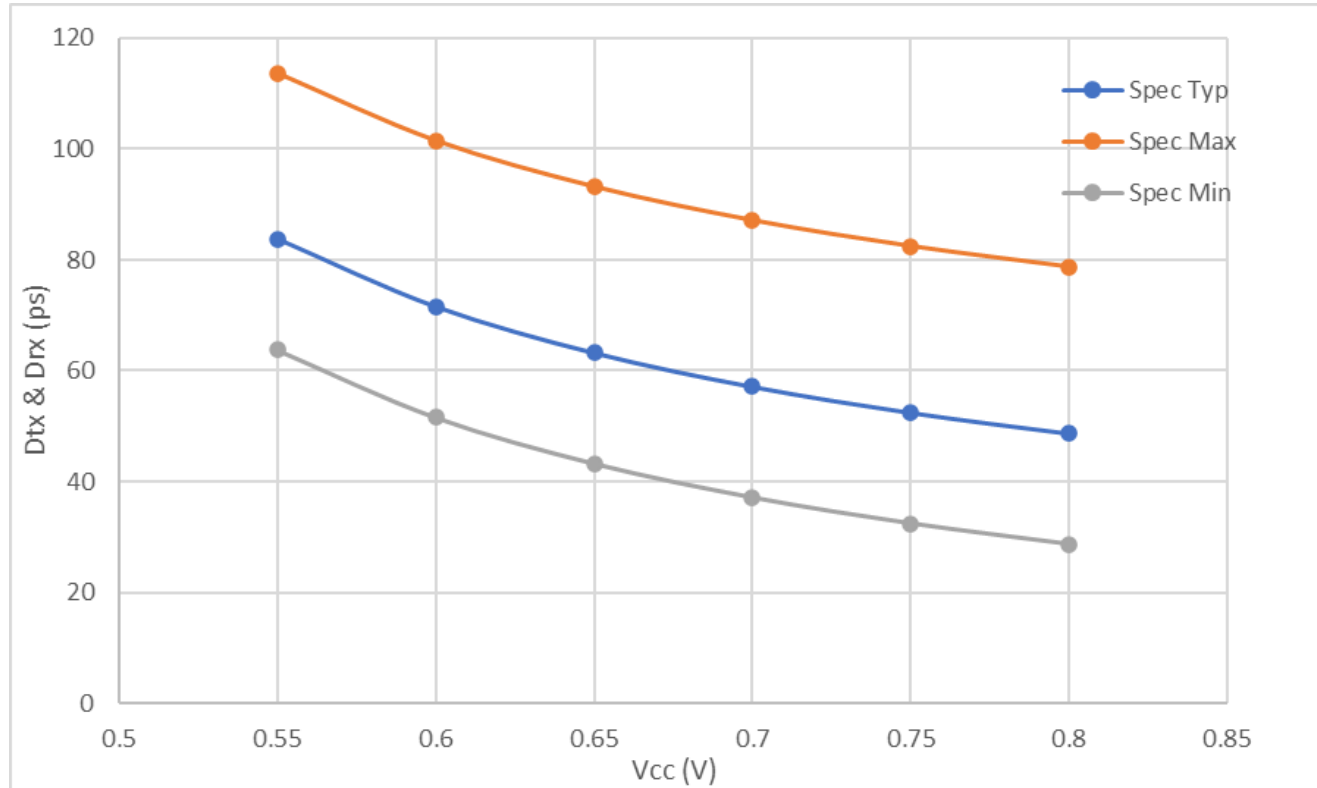
(2) Defined as clock to mean data, min/typ/max values are shown below.

(3) Alpha factor is defined as $\frac{dD_{tx}}{D_{tx}} / \frac{dV_{cc}}{V_{cc}}$ and $\frac{dD_{rx}}{D_{rx}} / \frac{dV_{cc}}{V_{cc}}$ for TX and RX respectively

(4) This is equivalent to a variation of +/- 5% in Vcc. Careful mitigation is particularly needed when disturbances external to UCIE occur, such as electromagnetic coupling from Through-Silicon Vias (TSVs).

Data/Clock Differential Delay Spec

Dtx and Drx Spec Range for 4Gb/s:



DVFS scenario, Dtx and Drx adjusts accordingly.

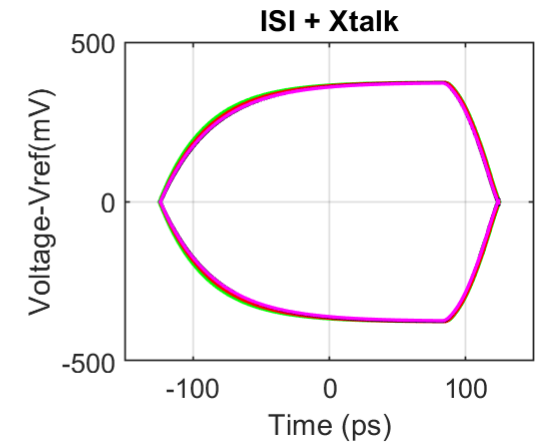
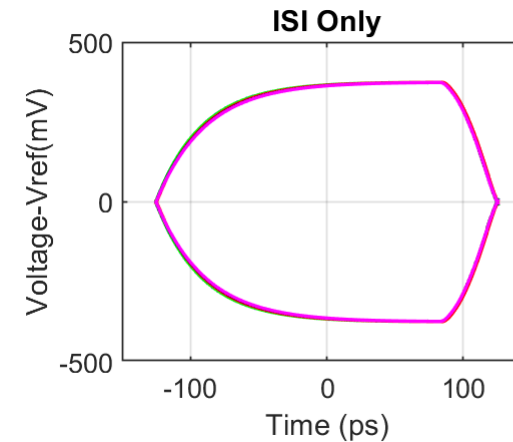
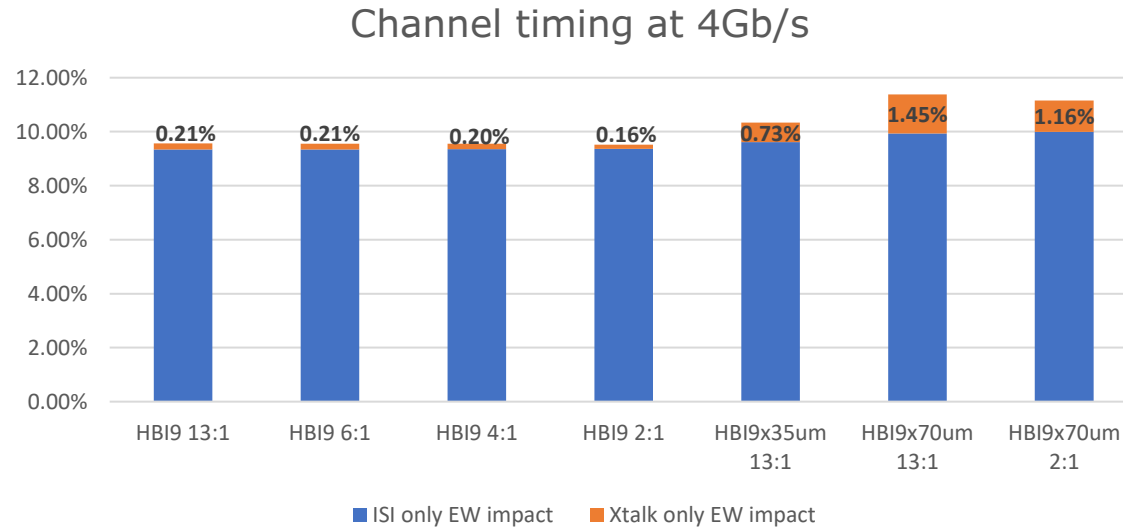
Delay doesn't need to conform to a fixed band across the entire Vcc range

Must satisfy equation:

$$C_h + J_{pw} + S_{tx} + S_{rx} + \sqrt{J_{rtx}^2 + J_{rrx}^2} + A_p + [\max(D_{tx}) - \min(D_{tx}) + \max(D_{rx}) - \min(D_{rx})](1 + \alpha_{trx}n_{vcc}) < 1 \text{ UI}$$

Example Channel Simulation

- Analysis shows feasibility of meeting Spec target



- Data rate 4 Gb/s
- Vdd = 0.75 V
- Rx EH = 250 mV
- TX Ron=350 Ohm
- TX Cap =20 fF
- RX Cap = 20 fF, unterminated

ESD Target

- 5V at introduction, scaling to 3V.
 - For sub 10μm
- Should explore the possibility of 0V for the special case of wafer-to-wafer hybrid bonding
 - Most dies are not exposed to ESD, and V tends to 0 as the gap approaches 0.
 - Need to consider test requirement and the center dies touched by handlers.

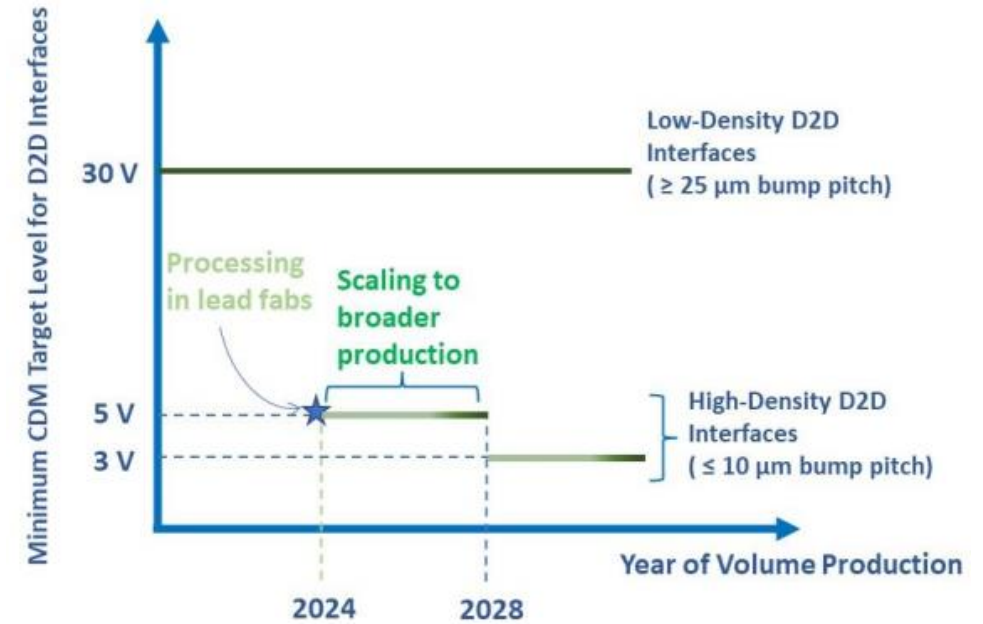
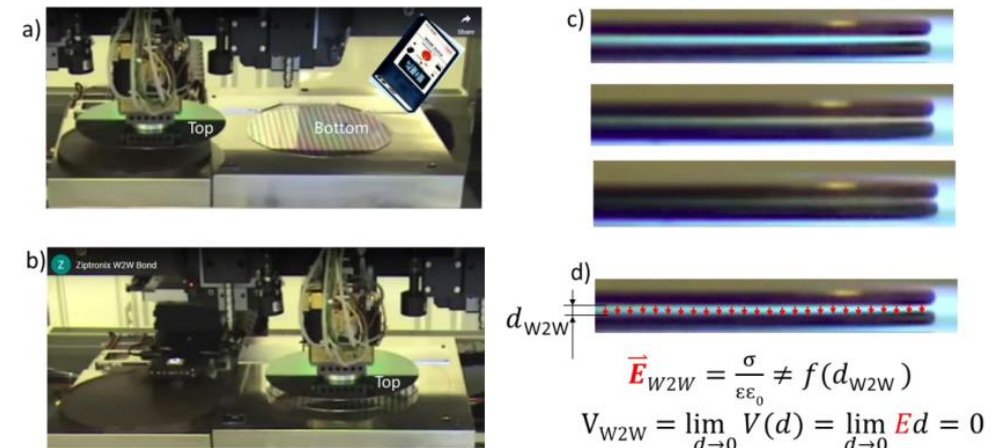


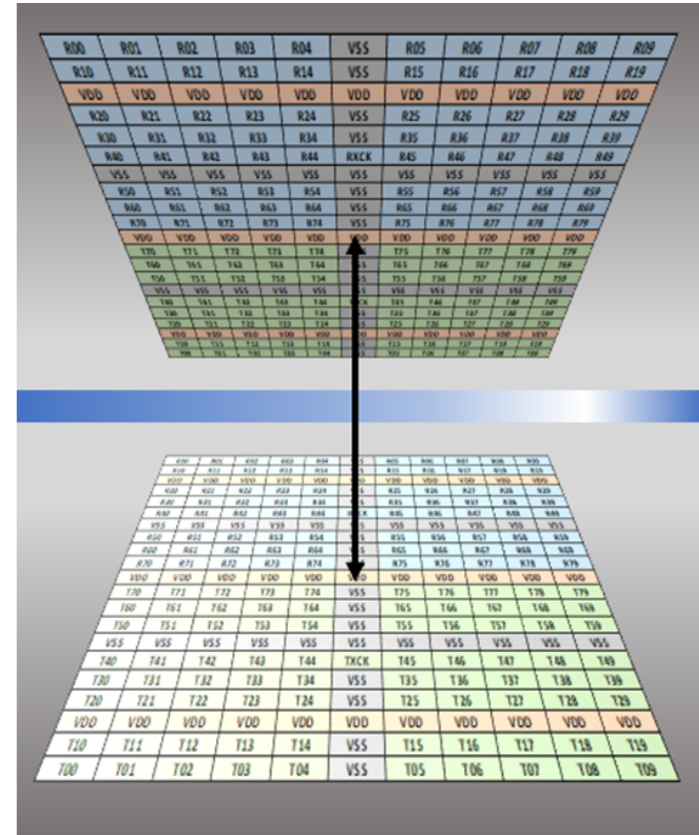
Figure 1: Roadmap of CDM Targets of Die-to-Die Interfaces



<http://www.jedec.org> reference JEP196
<http://www.esdindustrycouncil.org/ic/en> or <http://www.esda.org> reference White Paper 2 Part II

UCIe-3D Module Bump Map

21	R00	R01	R02	R03	R04	VSS	R05	R06	R07	R08	R09
20	R10	R11	R12	R13	R14	VSS	R15	R16	R17	R18	R19
19	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD
18	R20	R21	R22	R23	R24	VSS	R25	R26	R27	R28	R29
17	R30	R31	R32	R33	R34	VSS	R35	R36	R37	R38	R39
16	R40	R41	R42	R43	R44	RXCK	R45	R46	R47	R48	R49
15	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS
14	R50	R51	R52	R53	R54	VSS	R55	R56	R57	R58	R59
13	R60	R61	R62	R63	R64	VSS	R65	R66	R67	R68	R69
12	R70	R71	R72	R73	R74	VSS	R75	R76	R77	R78	R79
11	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD
10	T70	T71	T72	T73	T74	VSS	T75	T76	T77	T78	T79
9	T60	T61	T62	T63	T64	VSS	T65	T66	T67	T68	T69
8	T50	T51	T52	T53	T54	VSS	T55	T56	T57	T58	T59
7	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS
6	T40	T41	T42	T43	T44	TXCK	T45	T46	T47	T48	T49
5	T30	T31	T32	T33	T34	VSS	T35	T36	T37	T38	T39
4	T20	T21	T22	T23	T24	VSS	T25	T26	T27	T28	T29
3	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD	VDD
2	T10	T11	T12	T13	T14	VSS	T15	T16	T17	T18	T19
1	T00	T01	T02	T03	T04	VSS	T05	T06	T07	T08	T09
	1	2	3	4	5	6	7	8	9	10	11



- Arrange signals so that the same PHY may be used on top and bottom die
- Area Estimate: 0.02 mm² per x80 module (both TX and RX) at 9μm bump pitch. x70 module also defined in spec.
- No predefined adapter. Users have the flexibility to allocate some data lanes for adapter functions such as **Valid**, **Data Mask**, **Parity**, and **ECC**, as well as **Sideband**.

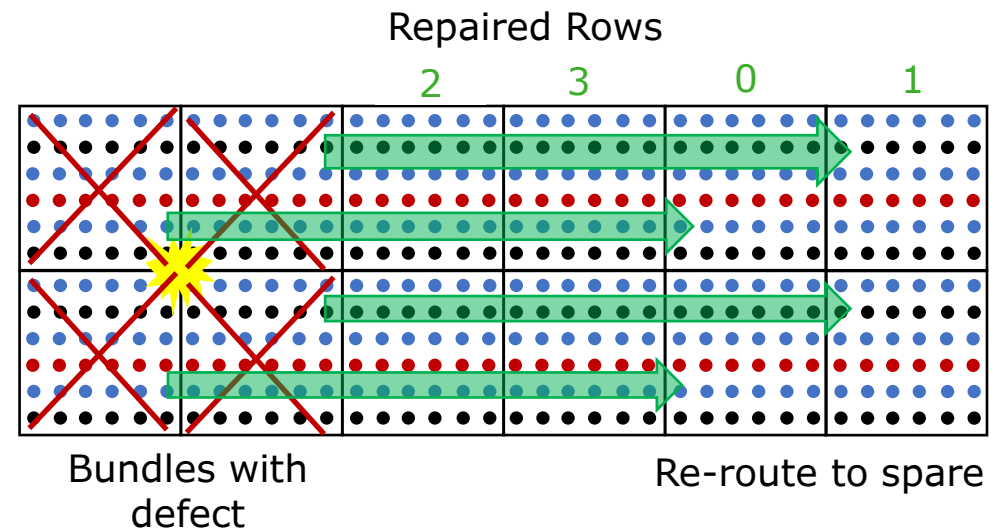
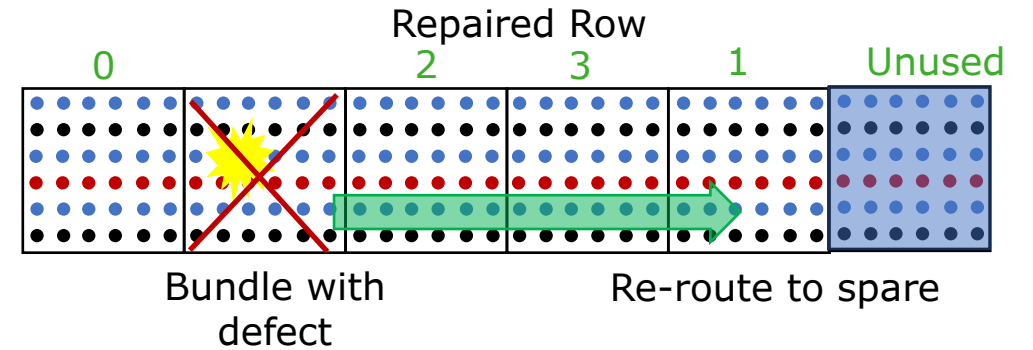
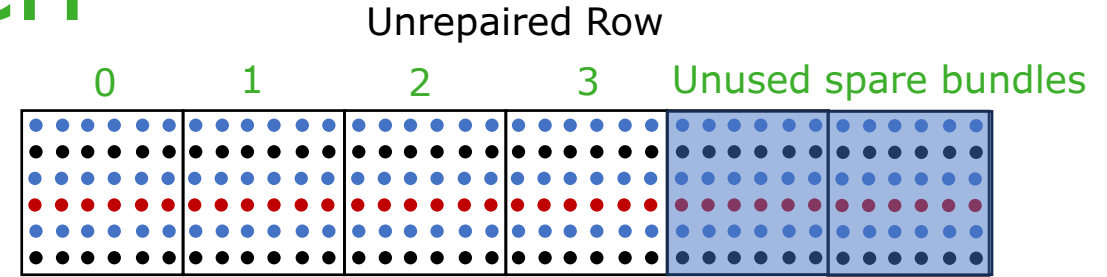
Bundle Repair Approach

- Reserve modules in SoC for repair, reroute to the backup module when there is a failure.
 - To address cluster failure mode. Defects are larger than bumps. For example, one defect can take out 5x5 bump area.
- UCIe-3D: Each Module has one TX bundle (x80 TX + Clock) and one RX bundle (x80 RX + Clock).
- For densely packed UCIe Module array, **reserve 2k full Modules**. k determined by:

$$1 - \sum_{i=0}^k P_i(AD_0) < \delta$$

$$P_i(x) = \frac{x^i}{i!} e^{-x}$$

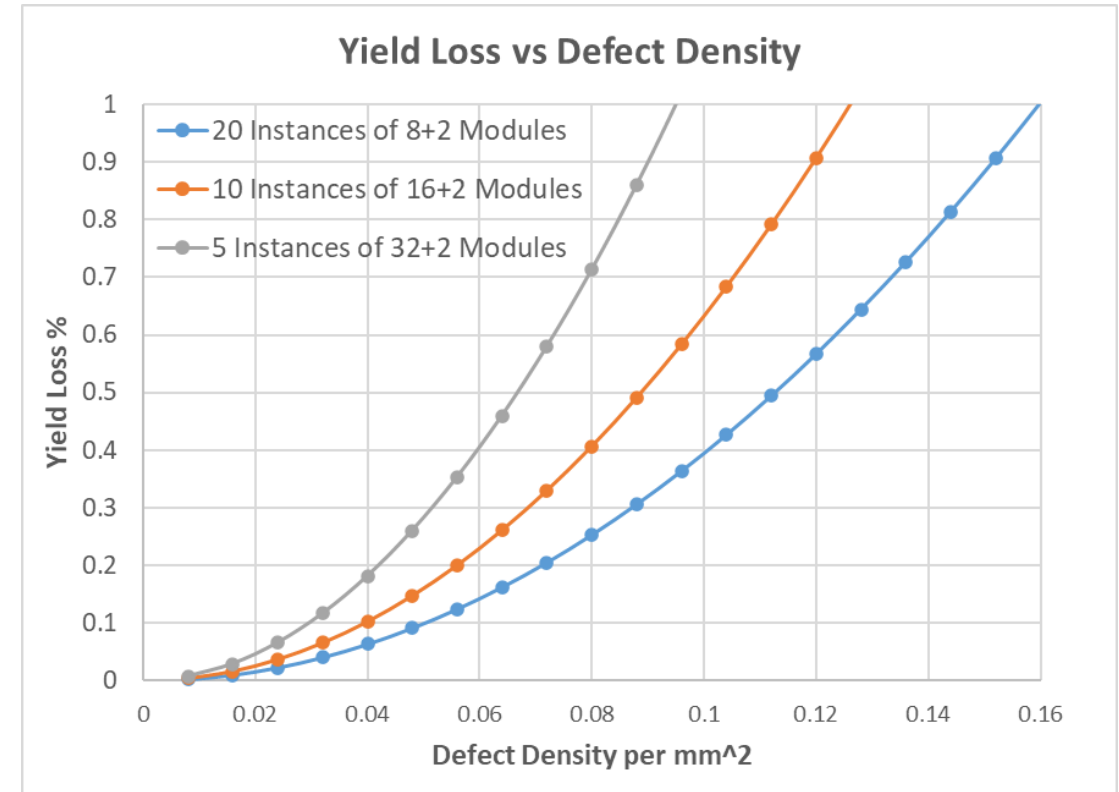
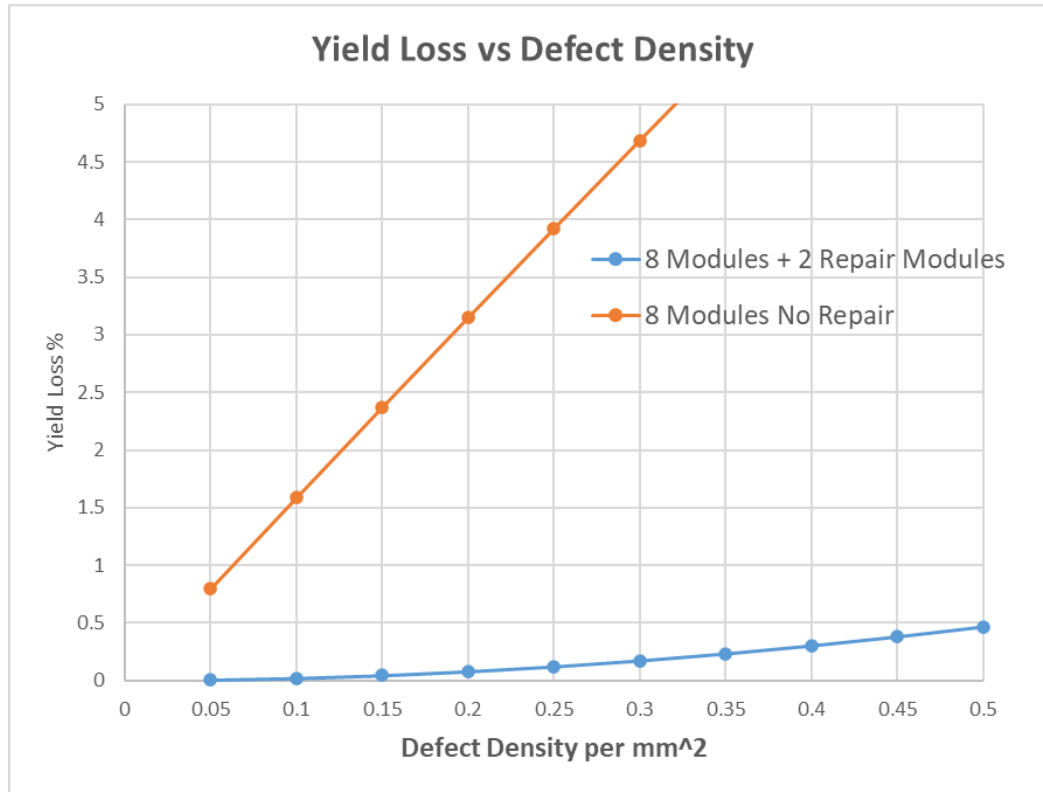
D_0 : defect density
 A : total UCIe-3D Si area
 δ : acceptable yield loss



Bundle Repair: Yield Estimate Examples

- 0.02mm² per module
- 8 Modules with 2 repairs vs no repair

- Total bi-directional BW 10.24TB/s
 - 20 instances of 8+2 Modules, area 4 mm².
 - 10 instances of 16+2 Modules, area 3.6 mm².
 - 5 instances of 32+2 Modules, area 3.4 mm².



UCIe-3D Performance Summary

Characteristics / KPIs	UCIe-S (2D)	UCIe-A (2.5D)	UCIe 3D	Comments for UCIe 3D
Characteristics				
Data Rate (GT/s)	4, 8, 12, 16, 24, 32		Up to 4	= SoC Logic frequency – power efficiency is critical
Width (each cluster)	16	64	80	Options or reduced width to 70, 60...
Bump Pitch (μm)	100 – 130	25 – 55	≤ 10 (optimized) $> 10 - 25$ (functional)	Must scale so that UCIe-3D fits within the bump area, must support hybrid bonding
Channel Reach (mm)	≤ 25	≤ 2	3D vertical	FtF bonding initially; FtB, BtB, multi-stack possible
Target for Key Metrics				
BW Shoreline (GB/s/mm)	28 – 224	165 – 1317	N/A (vertical)	
BW Density (GB/s/mm ²)	22 – 125	188 – 1350	4,000 – 300,000	4TB/s/mm ² @ 9 μm , ~12TB/s/mm ² @ 5 μm , ~35TB/s/mm ² @ 3 μm , ~300TB/s/mm ² @ 1 μm
Power Efficiency Target (pJ/b)	0.5	0.25	<0.05 at 9 μm -> 0.01 at 1 μm	Conservatively estimated at 9 μm pitch <0.02 for 3 μm pitch
Low-Power Entry/Exit	0.5nS $\leq$ 16G, 0.5-1nS $\geq$ 24G		0nS	No preamble or post-amble
Reliability (FIT)	0 < FIT (Failure in Time) << 1		0 < FIT << 1	BER < 1E-27
ESD	30V CDM		5V CDM $\rightarrow$ $\leq$ 3V	5V CDM at introduction, no ESD for W2W hybrid bonding possible

Questions

Thank You!

www.UCIexpress.org